

Jungseob Lee

NLP Researcher · Ph.D. Student, Korea University

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Summary

Integrated M.S./Ph.D. researcher at the Korea University NLP Lab (Advisor: Prof. Heuiseok Lim), specializing in large language models and natural language processing, with a focus on agentic LLMs, LLM efficiency, and efficient reasoning. First-author research has been published at **EMNLP 2026** (Main), **ACL 2025** (Oral), and **Findings of ACL 2024**. Four papers were accepted at **EMNLP 2026** (2 Main, 2 Findings), with co-authored work at **ICLR 2026** and a series of preprints under review. Recipient of the **Best Paper Award (Top 1)** at HCLT 2025. Served as the post-training Project Manager and Korea University representative for **VAETKI**, Korea's national Sovereign AI Foundation Model. Author of 30+ peer-reviewed papers across top-tier NLP venues and journals.

Research Interests: Large Language Models, Natural Language Processing, Agentic LLM, LLM Efficiency, Efficient Reasoning

Education

Integrated M.S./Ph.D. in Computer Science 2021 – Present
Korea University, Seoul, Korea **Expected: Feb. 2027**
NLP Lab, Advisor: Prof. Heuiseok Lim. Research focus: large language models, natural language processing, agentic LLM, LLM efficiency, and efficient reasoning.

Work Experience & Major Projects

VAETKI, Korean Sovereign AI Foundation Model Jun 2025 – Jan 2026
Post-training Project Manager, Korea University Representative NC AI (NCSOFT)-led consortium

- Served as post-training PM representing Korea University in Korea's national **Sovereign AI Foundation Model** program, organized by the Ministry of Science and ICT, an approximately KRW 200 billion national initiative with staged six-month evaluations running through 2027.
- The Korea University team (Prof. Heuiseok Lim) was selected as a core research partner in the NC AI (NCSOFT)-led grand consortium of 14 industry–academia–research institutions and 40 adopters (including POSCO DX, Lotte Innovate, KAIST, ETRI, and MBC).
- Led the full post-training pipeline for the VAETKI model, covering Supervised Fine-Tuning (SFT), Reinforcement Learning, and dataset curation.
- **VAETKI model:** an industry-specialized sovereign foundation model, open-sourced on Hugging Face on Dec 30, 2025. A 100B+ parameter Mixture-of-Experts architecture (∼11B active parameters at inference) with a custom Multi-head Latent Attention (MLA) reducing memory usage by up to ∼83%.
- Achieved state-of-the-art results on Korean language and culture (CLIcK), advanced reasoning (KoBALT), professional knowledge (KMMLU-Pro), and instruction following (IFEval).
- [Model] [Technical Report]
- Press: [Korea University] [AI Times] [Newstomato]

Cross-lingual Language Transfer for Large Language Models 2024 – 2025
Lead Researcher (first author) Published at ACL 2025 (Oral)

- Proposed a cross-lingual optimization method that efficiently transfers English-centric LLMs to new target languages while preserving source-language ability, accepted as an **Oral** presentation at ACL 2025.
- [PDF]

Efficient Reasoning for Large Reasoning Models 2025 – Present

Lead Researcher (first author) Findings of EMNLP 2026, preprints under review

- Leading a first-author line of work on training-free adaptive thinking budgets (DART) and stable efficiency training via adaptive correct-only rewards (ACOER) for large reasoning models. DART was accepted to **Findings of EMNLP 2026**.
- Co-first author of **ThinkFuse**, a trajectory-aware test-time fusion method for small reasoning models, accepted to **Findings of EMNLP 2026**.
- [DART] [ACOER] [Code]

Polyglot, Open-Source Multilingual Language Models 2022

Open-Source Contributor, EleutherAI

- Worked on efficient language transfer of pretrained language models with cross-lingual post-training (XPT) and MAD-X adapters, and also took part in pretraining.
- [GitHub]

Awards & Honors

Outstanding Reviewer Award 2026

The 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026), San Diego

Recognized by the ACL program committee for review quality and service to the conference. [Certificate]

Best Paper Award (Top 1) 2025

Annual Conference on Human and Cognitive Language Technology (HCLT 2025)

For *KULLM-RAG: A RAG-Specialized Large Language Model with Intelligent Self-Verification*, by **J. Lee** (first author), M. Kim, J. Yoon, S. Hong, Y. Jang, S. Lee, J. Seo, C. Park, J. Park, and H. Lim, from Korea University, Soongsil University, and Human-Inspired AI Research.

Minister of Science and ICT Award 2023

ICT Challenge 2023, Ministry of Science and ICT (organized by IITP)

Team of M. Kang, **J. Lee**, and S. Lee, for *KoCheckGPT*, Korean-specialized detection of LLM-generated documents for enterprise security, selected from 79 teams (263 participants) across 28 universities. [Press]

Publications

Underlined = me, * equal contribution. Full profile on Google Scholar.

EMNLP 2026 Main — **The Hallucination Signal Is a Mean Shift: Why Simple Probes Suffice**
J. Lee, J. Seo, H. Lim [arXiv]

EMNLP 2026 Main — **Language Chain in Alignment: Cross-Lingual Ranking Preference Optimization**
S. Lee, M. Kim, **J. Lee**, H. Lim [arXiv] [Code]

EMNLP 2026 Findings — **DART: Draft-Agreement Routing for Training-Free Adaptive Thinking Budgets in Hybrid Reasoning Models**
J. Lee, S. Hong, S. Lee, J. Seo, J. Son, S. Eo, C. Park, H. Park, H. Moon, H. Lim [arXiv] [Code]

EMNLP 2026 Findings — **ThinkFuse: Trajectory-Aware Test-Time Fusion for Small Reasoning Models**
M. Kang*, **J. Lee***, J. Seo, H. Lim [arXiv] [Code]

ICLR 2026 Poster — **Improving Semantic Proximity in Information Retrieval through Cross-Lingual Alignment**
S. Hong, Y. Jang, **J. Lee**, H. Moon, H. Lim [PDF]

KBS 2026 Journal — **Towards Enhancing Natural Language Inference for Identifying Erroneous Outputs in Data-to-Text Generation**
C. Ok, E.K. Lee, H. Ju, **J. Lee**, D. Oh *Knowledge-Based Systems* [DOI]

AACL 2026 Findings — **Skin-Deep: A Geometric Diagnostic for Alignment Fragility in Large Language Model Representations**
D.J. Lee*, **J. Lee***, S. Lee, S. Hong, S. Son, S. Eo, J. Seo, H. Lim [arXiv] [Code]

BIBM 2026 Short Paper

Structured Clinical Memory for Safety-Consistent Diagnosis and Recommendation Generation in Long Medical Dialogues
D.J. Lee, **J. Lee**, E. Park, T. Whang, H. Lim

ACL 2025 Oral

Cross-lingual Optimization for Language Transfer in Large Language Models
J. Lee, S. Hong, H. Moon, H.S. Lim [PDF]

HCLT 2025 Best Paper

KULLM-RAG: A RAG-Specialized Large Language Model with Intelligent Self-Verification
J. Lee, M. Kim, J. Yoon, S. Hong, Y. Jang, S. Lee, J. Seo, C. Park, J. Park, H. Lim *Annual Conference on Human and Cognitive Language Technology*

ESWA 2025 Journal

An Analysis on Language Transfer of Pre-trained Language Model with Cross-lingual Post-training
S. Son*, C. Park*, **J. Lee***, M. Shim, C. Lee, Y. Jang, J. Seo, J. Lim, H. Lim *Expert Systems with Applications* [DOI]

Report 2025 Technical

VAETKI Technical Report
NC AI Consortium [PDF]

ACL 2024 Findings

Length-aware Byte Pair Encoding for Mitigating Over-segmentation in Korean Machine Translation
J. Lee, H. Moon, S. Lee, C. Park, S. Eo, H. Ko, J. Seo, S. Lee, H.S. Lim [PDF]

Appl. Sci. 2024 Journal

Analysis of the Effectiveness of Model, Data, and User-Centric Approaches for Chat Application: A Case Study of BlenderBot 2.0
C. Park*, **J. Lee***, S. Son, K. Park, J. Jang, H. Lim *Applied Sciences* [DOI]

ACL 2023 Demo

PEEP-Talk: A Situational Dialogue-based Chatbot for English Education
S. Lee, Y. Jang, C. Park, **J. Lee**, J. Seo, H. Moon, S. Eo, S. Lee, B. Yahya, *et al.* [PDF]

Math. 2023 Journal

A Survey on Evaluation Metrics for Machine Translation
S. Lee*, **J. Lee***, H. Moon, C. Park, J. Seo, S. Eo, S. Koo, H. Lim *Mathematics* [PDF]

ESWA 2023 Journal

Doubts on the Reliability of Parallel Corpus Filtering
H. Moon, C. Park, S. Koo, **J. Lee**, S. Lee, J. Seo, S. Eo, Y. Jang, H. Kim, H. Lee, *et al.* *Expert Systems with Applications* [DOI]

HCLT 2023 Conference

KoCheckGPT: Korean LLM Written Document Detector
M. Kang, **J. Lee**, S. Lee, S. Hong, J. Park, H. Lim *Annual Conference on Human and Language Technology* [PDF]

COLING 2022 Conference

QUAK: A Synthetic Quality Estimation Dataset for Korean-English Neural Machine Translation
S. Eo, C. Park, H. Moon, J. Seo, G. Kim, **J. Lee**, H.S. Lim [PDF]

LREC 2022 Conference

Empirical Analysis of Noising Scheme based Synthetic Data Generation for Automatic Post-editing
H. Moon, C. Park, S. Lee, J. Seo, **J. Lee**, S. Eo, H.S. Lim [PDF]

Access 2022 Journal

K-NCT: Korean Neural Grammatical Error Correction Gold-standard Test Set Using Novel Error Type Classification Criteria
S. Koo, C. Park, J. Seo, S. Lee, H. Moon, **J. Lee**, H. Lim *IEEE Access* [PDF]

CCGPK 2022 Workshop

Focus on FoCus: Is FoCus Focused on Context, Knowledge and Persona?
S.Y. Lee*, **J. Lee***, C. Park, S. Eo, H. Moon, J. Seo, J. Park, H.S. Lim *1st Workshop on Customized Chat Grounding Persona and Knowledge* [PDF]

arXiv 2022 Preprint

There is No Rose Without a Thorn: Finding Weaknesses on BlenderBot 2.0
J. Lee, M. Shim, S. Son, C. Park, Y. Kim, H. Lim [arXiv]

arXiv 2022 Preprint

Language Chameleon: Transformation Analysis between Languages Using Cross-lingual Post-training
S. Son*, C. Park*, **J. Lee***, M. Shim, C. Lee, Y. Jang, J. Seo, H. Lim [arXiv]

JKCS 2022 Journal

Korean and Multilingual Language Models Study for Cross-lingual Post-training (XPT)
S. Son, C. Park, **J. Lee**, M. Shim, C. Lee, K. Park, H. Lim *Journal of the Korea Convergence Society* [PDF]

JKCS 2022 Journal

Policy-based Performance Comparison Study of Real-time Simultaneous Translation
J. Lee, H. Moon, C. Park, J. Seo, S. Eo, S. Lee, S. Koo, H. Lim *Journal of the Korea Convergence Society* [PDF]

HCLT 2022 Conference

SaJuTeller: Conditional Generation Deep-Learning based Fortune Telling Model
H. Moon, **J. Lee**, J. Seo, S. Eo, C. Park, W. Kim, J. Park, H. Lim *Annual Conference on Human and Language Technology* [PDF]

HCLT 2022 Conference

Korean Language Model Construction and Comparative Analysis with Cross-lingual Post-Training (XPT)
S. Son, C. Park, **J. Lee**, M. Shim, S. Lee, J.W. Lee, A. So, H. Lim *Annual Conference on Human and Language Technology* [PDF]

HCLT 2022 Conference

DART: Data Augmentation using Retrieval Technique
S. Lee, J. Seo, **J. Lee**, M. Kang, H. Moon, C. Park, D. Jung, J. Lee, K. Park, *et al.* *Annual Conference on Human and Language Technology* [PDF]

JKCS 2021 Journal

Empirical Study on BlenderBot 2.0's Errors Analysis in Terms of Model, Data and Dialogue
J. Lee, S. Son, M. Shim, Y. Kim, C. Park, A. So, J. Park, H. Lim *Journal of the Korea Convergence Society* [PDF]

Under Review

arXiv Oct 2026

Predicting and Repairing Merge Collapse in Large Language Models
J. Lee, S. Lee, S. Eo, H. Moon, J. Seo, H. Lim [arXiv] [Code]

arXiv Oct 2026

Distilling Directional Verification
J. Lee, S. Eo, S. Hong, S. Lee, C. Park, J. Seo, H. Lim [arXiv] [Code]

arXiv Oct 2026

CAST: Cost-Aware Speculative Trees from One-Pass Block Drafters
J. Lee, S. Eo [arXiv] [Code]

arXiv Oct 2026

Refusal Localizes, the Damage Relocates: Safety Layers Under Few-Sample Fine-Tuning
J. Lee*, D.J. Lee*, S. Eo, S. Hong, S. Lee, H. Lim [arXiv] [Code]

arXiv Sep 2026

Recovering Off-Policy Supervision for Speculative Decoding
J. Lee, C. Park, S. Eo, H. Moon [arXiv] [Code]

arXiv Sep 2026

Risk-Controlled Selective LLM Answering by Pricing Label-Free Checks
D.J. Lee*, **J. Lee***, C. Park, H. Moon, H. Lim [arXiv] [Code]

arXiv Sep 2026

Where Activation Sparsity and KV-Cache Sparsity Cross in LLM Decoding
J. Lee, S. Lee, S. Hong, S. Eo, H. Lim [arXiv] [Code]

arXiv Sep 2026

Faster Block-Diffusion Serving with Distribution-Free Risk Guarantees
J. Lee, D.J. Lee, C. Park, S. Eo, H. Lim [arXiv] [Code]

arXiv Sep 2026

Vision Is Not Overhead: One-Pass Block Drafting for Lossless Speculative Decoding in Vision-Language Models
J. Lee, S. Hong, D.J. Lee, C. Park, J. Seo, S. Eo, H. Lim [arXiv] [Code]

arXiv Jul 2026

Answer-Conditioned Chains of Thought Degrade Verifiable-Reasoning Distillation in Large Language Models
J. Lee, S. Lee, S. Son, D.J. Lee, S. Han, S. Eo, H. Lim [arXiv] [Code]

arXiv Jun 2026

To Isolate or to Score? Model-Adaptive Assessment for Cost-Efficient Multi-Agent RAG
J. Lee, C. Park, H. Lim [arXiv] [Code]

arXiv Jun 2026

Beyond Penalizing Mistakes: Stabilizing Efficiency Training in Large Reasoning Models via Adaptive Correct-Only Rewards
J. Lee*, S. Lee*, S. Hong, M. Kim, C. Park, H. Lim [arXiv] [Code]

arXiv Sep 2026

RenderRank: Learning to Rerank Text with Compressed Visual Tokens
S. Hong, Y. Jang, **J. Lee**, H. Moon, H. Lim [arXiv]

arXiv Jul 2026

LAMAR: An Open Language-Aware Multilingual Alignment Reranker
S. Hong, Y. Jang, **J. Lee**, S. Lee, H. Lim [arXiv]

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